Prune, Recover, Quantize: Compact MobileNetV2 for EuroSAT Classification

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Abstract. We compress the supplied distilled MobileNetV2 through residual-block removal, teacher-assisted recovery, and static INT8 quantization. The selected B3 model removes blocks 6, 9 and 16, recovers for three epochs at learning rate 0.001, then applies per-channel, 99.99-percentile quantization. It achieves 98.84% accuracy on the 27,000-image competition evaluation at 2.052 MiB. Compared with our previous quantization-only reference, it is 17.6% smaller with a 0.08-percentage-point accuracy tradeoff and improved worst-class accuracy. We distinguish measured accuracy and size from provisional local timing.

98.84%	**75.9% smaller**	**17.4% fewer**	**97.70%**
Overall accuracy	8.506 → 2.052 MiB	ONNX parameters	Worst-class accuracy

1. Task and evaluation protocol

We classify EuroSAT RGB images into ten land-use classes. The model accepts FP32 input `input`, shape $(1, 3, 224, 224)$, and returns ten scores. Preprocessing resizes images to 224, converts them to tensors, and applies ImageNet normalization. Export uses ONNX opset 13 with `dynamo=False`; inference uses ONNX Runtime 1.31.0 on CPU.

Per the recorded judge instruction, reported competition accuracy covers all 27,000 images, including recovery and calibration samples. A separate 21,600-image evaluation excludes recovery-training images, but is not an independent test of the supplied pretrained checkpoint. Local baseline values are our measurements, not organizer reference figures.

2. Compression pipeline and design choices

Pipeline. Supplied checkpoint → remove three residual blocks → adaptive BatchNorm → knowledge-distillation recovery → FP32 ONNX → graph preprocessing → recalibration → static INT8 QDQ.

Pruning exploration. We screened 26 Stage A records on 2,700 fixed images, disjoint from the 5,400 recovery images (seed 0). Width sweeps used Torch-Pruning 1.6.1, magnitude importance, global/isomorphic or uniform channel selection, multiples of eight, and a protected classifier. At 10% global width pruning, accuracy fell from 99.37% to 77.19% without recovery. Alternative BN-scale and Taylor criteria did not rescue the 30% setting.

Depth pruning was less damaging. Replacing residual blocks 6, 9 or 16 individually with Identity gave 98.96%, 98.96% and 98.93%. Removing all three gave 97.56%; four and five removals fell to 94.07% and 91.07%. Each candidate used adaptive BN (100 batches of 32). We selected the three-block structure; **the final model uses no width pruning**.

Recovery. The initial one-epoch CPU attempt at learning rate 0.005 achieved only 96.52% after INT8. The selected run restarts from the three-block pruned model and uses the recipe below. Both learning rate and duration changed, so their individual effects are not isolated.

Table 1: Selected recovery recipe: `b3_kd_lr001_e3`.

Setting	Value
Recovery data / seed	5,400 images / 0
Teacher / supervision	ResNet-50 / labels + cached logits
Epochs / batch / augmentation	3 / 32 / none
SGD learning rate / schedule	0.001 / cosine to zero
Momentum / weight decay	$0.9 / 4 \times 10^{-5}$
KD weight / temperature	$\alpha = 0.7 / T = 4$
Training device	Apple M3 Pro GPU (MPS)
Recorded epoch time	31 + 28 + 28 = 87 seconds

The loss combines 0.3 CE with $0.7T^2$ times teacher-to-student KL divergence. Epoch times exclude setup and evaluation. Quick-eval accuracy progressed through 97.41%, 99.26% and 99.22%; the submitted model uses epoch three. Its pre-quantization accuracy on the recovery-excluded 80% was 98.75%.

Quantization. We run `quant_pre_process`, then static QDQ with signed INT8 weights and activations (S8S8), per-channel weight scales, percentile 99.99, and no excluded layers. Activation ranges are recalibrated on the recovered model using 300 fixed images (30/class, seed 42), rather than reused from the unpruned model [1].

Table 2: Calibration study on the unpruned model.

Configuration	Acc. %	MiB
Per-tensor MinMax, U8S8	96.32	2.277
Per-channel MinMax, U8S8 / S8S8	98.09	2.491
Per-channel entropy	98.09	2.491
Q3a: percentile 99.99	98.92	2.491
Percentile 99.999	98.69	2.491
Q3a, first/last layers FP32	98.86	2.528
FP16	99.17	4.285

Per-channel scales gained 1.77 points at fixed U8S8; percentile calibration gained a further 0.83 points over per-channel MinMax. Calibration with 100 or 1,000 images scored 98.83% or 98.90%. These experiments motivated the 300-image percentile recipe reused after pruning.

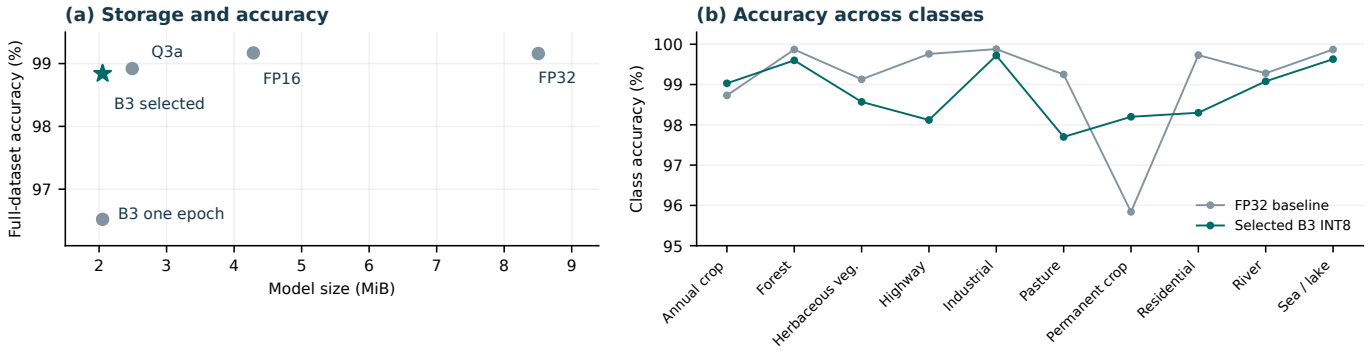


Figure 1: Full-dataset accuracy versus model size (left), and per-class accuracy for the FP32 baseline and selected B3 INT8 (right). Pruning plus recovery changes the class tradeoff: PermanentCrop improves, while several other classes decline.

3. Final results and tradeoffs

Table 3: Submission versus FP32 and prior Q3a reference.

Metric	FP32	Q3a	B3 INT8
Overall accuracy (%)	99.16	98.92	98.84
Worst-class accuracy (%)	95.84	95.24	97.70
File size (MiB)	8.506	2.491	2.052
ONNX parameters (M)	2.2196	2.2196	1.8338
ONNX MACs (G)	0.2995	0.2995	0.2627
Median latency (ms)	2.554	0.836	1.110*
Peak process RSS (MiB)	87.91	76.86	77.11*
Estimated energy (J)	0.05993	0.01938	0.04989*

*B3 local timing is provisional; FP32/Q3a use the earlier five-round milestone. These are not matched-session speed comparisons.

The selected model is 4.15× smaller than FP32, with 17.4% fewer parameters and 12.3% fewer MACs. Overall accuracy drops 0.32 percentage points. Against Q3a, it saves another 17.6% of storage for 0.08 points of accuracy. Its weakest class is Pasture (97.70%); PermanentCrop improves from 95.84% in FP32 to 98.20%. Compared with FP32, Highway and Residential fall by 1.64 and 1.43 points, respectively. Thus the gain in worst-class accuracy does not mean every class improves.

The recovered FP32 and INT8 models both score 98.84% overall at reported precision. Quantization reduces the recovered model from 7.028 to 2.052 MiB with no rounded overall-accuracy loss. Selection favors the smaller, more balanced model despite being 0.06 points below the original 98.9% planning target; this tradeoff is explicit.

Local measurement limits. Inference uses an 11-core M3 Pro with 18 GB RAM, batch one, and default ORT threads. The selected run used ten warm-ups and 100 timed inferences; background CPU was 48.4%. A new five-round benchmark refused to start at 43% CPU on battery. We therefore do not claim a verified local speedup over Q3a. Energy is a CodeCarbon CPU-load/TDP estimate over 15 seconds, not a power-sensor measurement.

4. Qualcomm device profile

We use the Dragonwing RB3 Gen 2 Vision Kit (QCS6490, Qualcomm Linux 1.6), selected after the project's catalog check found no UNO Q / QRB2210 profile. This is a listed Linux IoT proxy, not measured UNO Q performance. A new profile of the selected B3 reports **5.523 ms**, versus 26.690 ms for FP32 (4.83×). Estimated peak memory is 27.06 MiB versus 28.10 MiB for FP32. Both profiles execute on CPU; no NPU acceleration is claimed. Compile/profile IDs are `jgn1kl6vp`/`jp2oe0kxg`. The older Q3a profile is retained only as a separate reference.

5. Lessons and reproducibility

The first recovery attempt showed that removing blocks alone was insufficient. The later three-epoch, lower-learning-rate run substantially recovered accuracy, and MPS reduced recorded training-loop time to 87 seconds. This is evidence for the combined recipe, not proof that either change alone caused the improvement. Width pruning, teacher-versus-label-only recovery, quantization-aware training and best-epoch selection remain opportunities for further controlled study.

The submitted ONNX file is byte-identical to the selected three-epoch B3 artifact, SHA-256 prefix `e80fd982b0ab1c8f`. Its graph has 43 convolutions and seven residual additions (versus 52 and ten before pruning), with 54 QuantizeLinear and 142 DequantizeLinear nodes. The saved checks confirm the required input/output shapes and absence of dynamic-quantization operators.

The source bundle includes fixed subset indices, pruning and recovery configurations, training/export code, quantization settings and raw results. Run `reproduce_selected.sh` in a fresh workspace with the supplied student/teacher checkpoints and EuroSAT data; it performs pruning, teacher caching, recovery, export and quantization. The original run used MPS; CPU reproduction may differ numerically. `metrics.json` records the complete model hash and measurement provenance. The 98.75% recovery-excluded score belongs to the FP32 recovered model; no separate INT8 score on that subset is claimed.

References

[1] ONNX Runtime. *Quantize ONNX models*. Documentation, accessed October 2026.

Appendix A. Dashboard data at a glance

Recorded experiments from 13 October 2026. Values below come directly from the JSON files behind the dashboard; archived and smoke-test runs are excluded. Teal highlights the submitted model. The first two pages remain the main technical report.

A1. Full-dataset experiments: all 27,000 EuroSAT images

Run	Acc. %	Worst %	Params M	MACs G	Size MiB	Latency ms	RAM MiB	Energy J/image	Time basis
FP32 ONNX	99.16	95.84	2.220	0.2995	8.506	2.554	87.91	0.05993	M
FP32 PyTorch	99.16	95.84	2.237	0.3262	8.772	19.862	303.19	0.32409	S
Q0	96.32	88.13	2.220	0.2995	2.277	0.905	75.56	0.02251	P
Q1	98.09	95.45	2.220	0.2995	2.491	0.824	76.42	0.01843	P
Q1b	98.09	95.45	2.220	0.2995	2.491	0.928	76.42	0.05401	P
Q2	98.09	95.45	2.220	0.2995	2.491	0.835	77.17	0.02076	P
Q3a: quant-only	98.92	95.24	2.220	0.2995	2.491	0.836	76.86	0.01938	M
Q3b	98.69	95.92	2.220	0.2995	2.491	0.836	77.08	0.01809	P
Q4	98.86	94.88	2.220	0.2995	2.528	1.056	77.84	0.02448	P
Q6a: 100 images	98.83	94.72	2.220	0.2995	2.491	0.912	77.16	0.11928	P
Q6b: 1,000 images	98.90	94.96	2.220	0.2995	2.491	0.838	78.52	0.03317	P
Q7: FP16	99.17	95.88	2.220	0.2995	4.285	2.708	89.33	0.06280	M
B3: 1-epoch FP32	96.49	88.52	1.834	0.2627	7.028	2.263	83.25	0.06386	P
B3: 1-epoch INT8	96.52	89.28	1.834	0.2627	2.052	0.740	75.22	0.01599	P
B3: 3-epoch FP32	98.84	97.60	1.834	0.2627	7.028	2.302	84.27	0.06285	P
B3: 3-epoch INT8 (selected)	**98.84**	**97.70**	**1.834**	**0.2627**	**2.052**	**1.110**	**77.11**	**0.04989**	**P**

Reading the table. M = median of five milestone rounds; P = provisional single-session measurement; S = single-session reference. Latency is the median within a round. All energy values are CodeCarbon estimates; unusually high provisional values should not be treated as settled comparisons. Worst class is Forest for Q0, Pasture for Q1/Q1b/Q2 and the three-epoch B3 runs, and PermanentCrop otherwise. ONNX and PyTorch counts/runtime overhead differ; compare within the same runtime. Q0–Q7 are quantization-only variants. B3 removes blocks 6, 9 and 16; the selected run uses three recovery epochs at learning rate 0.001. Its local timing remains provisional.

A2. Pruning exploration: 2,700-image quick evaluation, before recovery training

All candidates use adaptive BN before evaluation. Counts are measured with the same PyTorch/thop method as the reference. These subset accuracies are not directly comparable with A1; latency, RAM and energy were not measured for these Stage A runs.

Candidate	Acc. %	Worst %	Params M	MACs G	MiB
Reference + BN	99.37	97.60	2.237	0.3262	8.506
Drop block 3	97.33	91.20	2.228	0.2966	8.472
Drop block 5	98.52	93.60	2.222	0.3139	8.449
Drop block 6	98.96	95.60	2.222	0.3139	8.449
Drop block 8	98.67	96.00	2.182	0.3153	8.300
Drop block 9	98.96	95.20	2.182	0.3153	8.300
Drop block 10	98.48	93.20	2.182	0.3153	8.300
Drop block 12	98.48	93.20	2.118	0.3025	8.058
Drop block 13	98.41	95.60	2.118	0.3025	8.058
Drop block 15	98.44	94.00	1.917	0.3103	7.291
Drop block 16	98.93	96.00	1.917	0.3103	7.291
Stack 1 block	98.96	95.60	2.222	0.3139	8.449
Stack 2 blocks	98.52	93.20	2.168	0.3030	8.243
Stack 3 blocks	97.56	92.00	1.848	0.2871	7.028
Stack 4 blocks	94.07	84.40	1.793	0.2761	6.822
Stack 5 blocks	91.07	79.60	1.778	0.2638	6.764

Candidate	Acc. %	Worst %	Params M	MACs G	MiB
Global 10%	77.19	47.20	1.808	0.2247	6.878
Global 20%	57.19	16.67	1.439	0.1789	5.476
Global 30%	39.41	5.67	1.095	0.1421	4.170
Global 40%	18.85	0.00	0.828	0.1099	3.157
Uniform 10%	76.26	43.20	1.784	0.2366	6.788
Uniform 20%	65.00	51.20	1.421	0.1984	5.409
Uniform 30%	39.15	27.00	1.092	0.1527	4.158
Uniform 40%	21.37	0.00	0.810	0.1144	3.089
30% BN-scale	15.81	0.00	1.025	0.1584	3.903
30% Taylor	22.74	0.00	1.003	0.1556	3.819

Depth-stack key. Blocks are added in this order: 6, 9, 16, 8, 5. Thus “Stack 3” removes 6+9+16; “Stack 4” also removes block 8.

Width settings. Global and uniform sweeps use magnitude importance. The requested percentage is a channel-pruning setting, not a percentage of total weights. BN-scale and Taylor are alternative importance criteria at 30%.

Why B3 was selected. Three-epoch recovery raises INT8 accuracy to 98.84%. It is 17.6% smaller than Q3a for a 0.08-point overall-accuracy tradeoff, with a 97.70% worst-class score. The one-epoch attempt is retained as historical evidence.

Source and precision. A1 uses `results/*.json`, replacing timing fields with `milestone_timing` when present, as the dashboard does. A2 uses `results/stageA/*.json`. Figures are rounded only for display; the saved JSON records retain the measurement precision. M = million parameters; G = billion multiply–accumulate operations; MiB = 2^{20} bytes.